

Airbnb NYC Supply-Demand

Business Intelligence Project Report

Overall Analysis & BI Query Responses

Team Members	Ayesha Salahuddin (29157) Narin Nadeem (29167)
Dataset	NYC Airbnb Open Data from Kaggle
BI Tool	Microsoft Power BI Desktop
Programming	Python
Stakeholder	Regional Operations Manager - Airbnb NYC
Domain Expert	Mohammad Shahmeer - Asst. Manager, Livedin

1. Dataset Description & Industry Background

1.1 About the Dataset

The dataset used in this project is the New York City Airbnb Open Data, sourced from Kaggle (uploaded by Arian Azmoudeh). It contains listing activity data for Airbnb properties in New York City and includes full descriptions, average review scores, host information, pricing data, and availability details for each listing.

Total Rows	102,599 listings (101,857 after cleaning)
Total Columns	26 original columns (24 after cleaning)
Geographic Scope	New York City, 5 Boroughs (Manhattan, Brooklyn, Queens, Bronx, Staten Island)
Key Facts	price, service fee, availability 365, number of reviews, reviews per month, review rate number, minimum nights, calculated host listings count
Key Dimensions	neighbourhood group (borough), neighbourhood, room type, host_identity_verified, cancellation_policy, instant_bookable, Construction year
Time Coverage	Listings with last review dates ranging from 2015–2022
Data Source	https://www.kaggle.com/datasets/arianazmoudeh/airbnbopendata

1.2 Industry Background

Airbnb, Inc. is an American company founded in 2008 that operates an online marketplace for lodging, primarily homestays for vacation rentals and tourism activities. The company is based in San Francisco, California, and the platform is accessible via website and mobile app. Airbnb does not own any of the listed properties, instead, it profits by receiving commission from each booking, typically charging hosts around 3% and guests up to 14.2% per transaction.

Since 2008, guests and hosts have used Airbnb to travel in a more unique, personalized way. As part of the Airbnb Inside initiative, this dataset describes the listing activity of homestays specifically in New York City, one of Airbnb's highest-demand global markets. NYC is also one of the most regulated short-term rental markets in the world, with Local Law 18 significantly restricting the number of active listings, creating an even more acute supply-demand imbalance.

The short-term rental industry operates on a two-sided marketplace model: hosts provide supply (available listings) while guests generate demand (bookings and searches). The core challenge for operations managers is ensuring that supply keeps pace with demand in real time particularly during peak seasons, local events, and holidays when demand spikes unpredictably.

Business Problem: A Regional Operations Manager at Airbnb needs a way to identify and predict geographic supply-demand gaps in real time because relying on lagging historical data makes it difficult to recruit hosts fast enough to meet sudden surges in guest interest.

2. Data Wrangling & EDA Summary

2.1 Data Validation Findings

Before cleaning, the dataset was validated for data types, null values, duplicates, and unexpected values. The following key issues were identified:

- price and service fee columns stored as strings with \$ prefix, required type conversion
- neighbourhood group contained typos: 'brookln' and 'manhatan' data entry errors
- minimum nights had invalid negative values (min: -1,223) and extreme outliers (max: 5,645)
- availability 365 had negative values (min: -10) and impossible values exceeding 365 (max: 3,677)
- license column had 102,597 nulls out of 102,599 rows effectively empty
- house_rules had 52,131 nulls (51%) and was unstructured free text
- last review stored as string required datetime conversion
- 15,893 nulls in last review and reviews per month representing listings never reviewed

2.2 Data Cleaning Steps

The following 10 cleaning steps were applied systematically:

#	Step	Action Taken
1	Drop Irrelevant Columns	Dropped license, house_rules, country, country code, NAME, host name. 6 columns removed
2	Fix Borough Typos	Corrected 'brookln'→Brooklyn, 'manhatan'→Manhattan; dropped 29 null borough rows
3	Clean Price & Service Fee	Stripped \$ and commas, converted to float; dropped null/zero price rows (~247 removed)
4	Fix Minimum Nights	Removed negative values, capped at 365, filled 409 nulls with median
5	Fix Availability 365	Removed negatives, capped at 365, filled 448 nulls with 0
6	Fill Categorical Nulls	host_identity_verified→'unconfirmed' (289), cancellation_policy→'moderate' (76), instant_bookable→False (105)
7	Handle Review Columns	number of reviews and reviews per month nulls filled with 0; review rate number nulls filled with median
8	Convert Last Review	Converted string to datetime; derived days_since_last_review feature (9,999 for never-reviewed listings)
9	Drop Null Coordinates	Dropped 8 rows with null lat/long; dropped 15 rows with null neighbourhood
10	Derive New Features	Created is_inactive, is_at_risk, and host_type columns for BI analysis

Final cleaned dataset: 101,857 rows × 24 columns. Original: 102,599 rows × 26 columns. Total rows removed: 742.

2.3 Descriptive Statistics - Consolidated Results

Variable	Mean	Median	Std Dev	Min	Max	Skewness
price (USD)	\$625.37	\$624.00	\$220.41	\$50	\$1,200	0.02
service fee (USD)	\$125.16	\$124.00	\$44.21	\$10	\$240	0.02
minimum nights	7.81	3.00	19.45	1	365	8.54
number of reviews	29.14	7.00	55.43	0	1,024	5.12
reviews per month	1.16	0.74	1.66	0	58.5	7.34
review rate number	3.28	3.00	1.02	1	5	0.11
host listing count	5.74	1.00	29.85	1	327	8.91
availability 365	140.26	96.00	131.11	0	365	0.72

2.4 Key EDA Findings

- Manhattan and Brooklyn dominate listings: together accounting for approximately 83% of all NYC Airbnb supply
- Entire home/apt listings (52.3%) slightly outnumber Private rooms (45.3%); Shared rooms and Hotel rooms together make up less than 2.3%
- Price is uniformly distributed between \$50–\$1,200 with very low skewness (0.02): no extreme outliers after cleaning
- Number of reviews is highly right-skewed (5.12), most listings have very few reviews while a small number have hundreds
- Availability 365 mean of 140 days vs median of 96 days indicates many listings have limited availability, suggesting either high occupancy or inactive listings
- 7,000 listings flagged as at-risk (reviews/month < 0.5, rating < 3.0, availability < 30 days, no review in 180+ days)
- 4,000 listings flagged as inactive (0 reviews + availability > 300 days)
- ANOVA test confirmed nightly prices differ significantly across NYC boroughs ($p < 0.05$)
- Pearson correlation between price and rating is near-zero, guests do not pay more for higher-rated listings

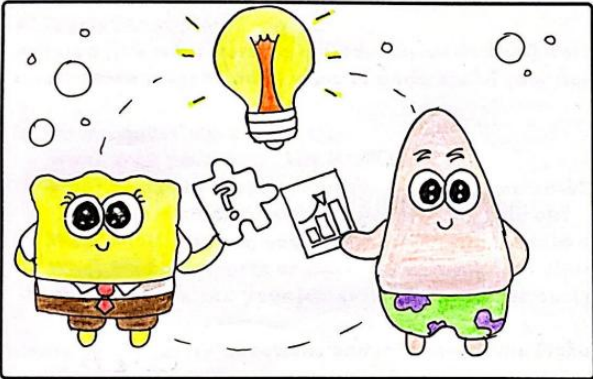
3. Design Thinking Framework Implementation

3.1 Overview

This project followed the five-stage Design Thinking framework for Business Intelligence: Empathize, Define, Ideate, Prototype, and Test. The design sprint was completed as a team using a structured workbook, and the insights from each stage directly shaped the BI solution developed.

YOUR BI TEAM

Create a doodle of your team - be creative!



What's your dataset?

New York City Air BnB Data :
It includes full descriptions and Average review score and details of the Airbnb availability for the day.

Potential Facts

Price, service fee, minimum night, numberOfReviews, reviews per month, review rate number, availability 365, calculated host listings count.

Potential Dimensions

Dim Host, Dim Location, Dim Listing Type, Dim Policy, Dim Time.

3.2 Stage 1 - Empathize

The Empathize stage involved two activities: a simulated AI domain expert interview during the design sprint, and a formal interview with a real business professional, Mohammad Shahmeer (Assistant Manager – Business Excellence at Livedin), whose company operates in the short-term rental industry.

Design Sprint Empathy Map

The design sprint empathy map captured the perspective of a Regional Operations Manager at Airbnb. Key findings:

- **THINK & FEEL:** Frustrated that data arrives too late to act; anxious about sudden demand surges; pressured to recruit hosts faster than tools allow
- **SEE:** Weekly summary reports with lagging data; manually maintained event calendars; dashboards showing post-damage rating scores
- **HEAR:** Reviews and ratings, marketing campaigns, host complaints about pricing gaps, host success team alerts

- SAY & DO: 'By the time I see data, it's too late'; manually cross-references event calendars with booking data
- PAINS: Lagging data makes real-time decisions impossible; no automated alert system for at-risk listings
- GAINS: Real-time geo-demand alerting; early warning dashboard; fast host recruitment triggered by demand forecasts

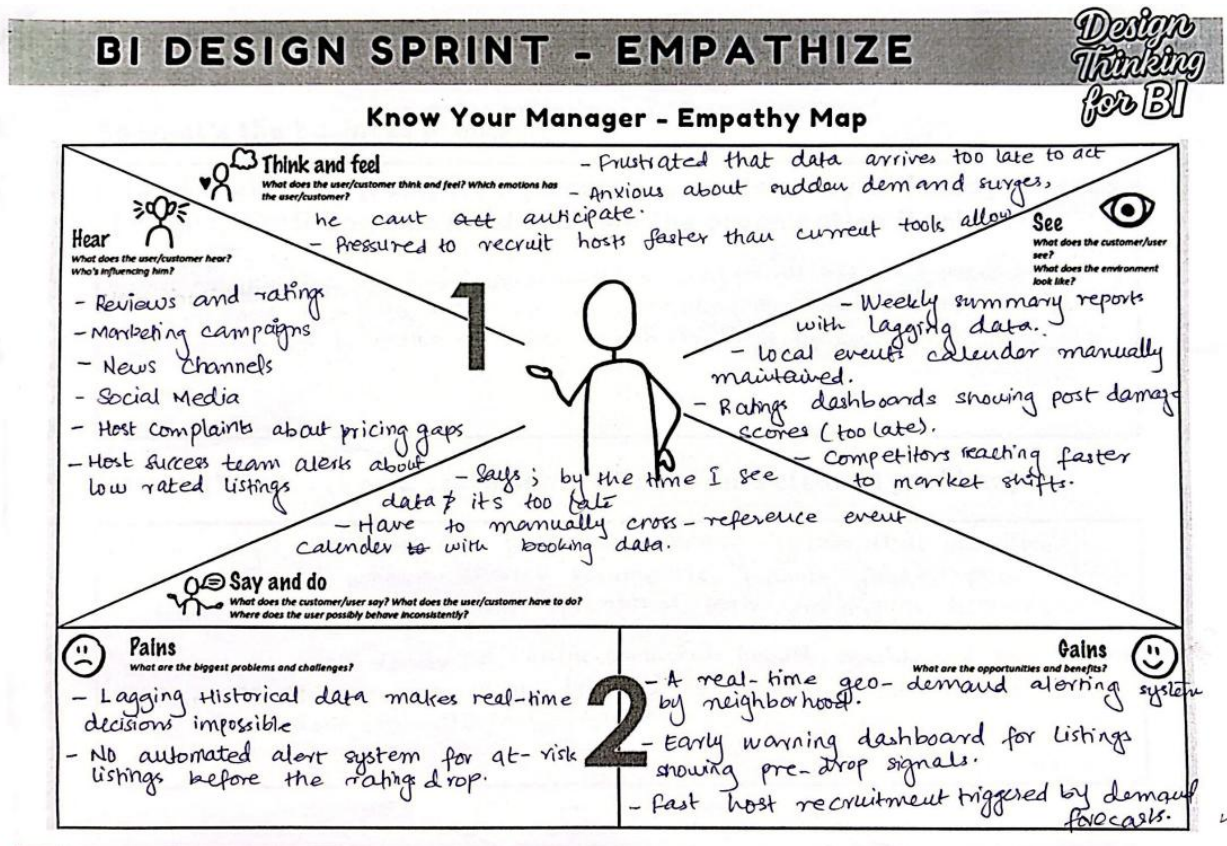


Figure: Hand-drawn empathy map from the design sprint workbook showing the Regional Operations Manager's perspective across all six quadrants.

Domain Expert Interview - Mohammad Shahmeer

Mohammad Shahmeer serves as Assistant Manager – Business Excellence at Livedin, a short-term rental and property management company operating in Pakistan, closely analogous to Airbnb. The interview was conducted to validate our problem statement and gather real-world operational insights.

Full Name	Mohammad Shahmeer
Role	Assistant Manager – Business Excellence
Company	Livedin
LinkedIn	linkedin.com/in/mshahr
Interview Format	Audio recording via WhatsApp

Key Insights from Interview

- Supply cannot respond to demand in real time; getting a new property live takes more than a few days, validating our problem statement about slow host recruitment
- Ratings, reviews, cancellation rates, and on-ground audits are primary at-risk signals — aligning with Dashboard 2 design
- Current tools (Amazon QuickSight + SQL + PriceLabs) provide market-level data but lack granularity into the customer journey
- Demand is strongly seasonal (October–February peak in their market), confirming need for time-based trend analysis
- AI and automation are already being embraced automated alerts in our dashboard align with the stakeholder's direction

Interviewee Mohammad Shahmeer	Role Asst. Manager – Business Excellence	Company Livedin	LinkedIn linkedin.com/in/mshahr
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THINK & FEEL

What does the stakeholder think and feel? What emotions came through?

- Feels pressure to constantly improve systems and processes across the business
- Concerned when demand spikes during peak season (Oct–Feb) and supply cannot keep up
- Frustrated when data is available but not granular enough to extract deep insights
- Motivated by leveraging AI and automation to solve problems the team previously found difficult
- Anxious about lead conversion rates feels the funnel performance is a persistent challenge
- Values data-driven decision making but feels current tools don't go deep enough into the customer journey

HEAR

What do they hear from colleagues, the market, or guests?

- Hears from different business heads about operational inefficiencies and improvement needs
- Receives guest feedback through integrated chat systems and meeting call logs
- Hears about competitor listings and market pricing signals through PriceLabs
- Gets alerts when ratings or reviews indicate a listing is underperforming
- Hears from the sales team about low lead-to-booking conversion rates
- Receives property condition feedback from the on-ground audit team

SEE

What does the stakeholder see in their work environment?

- Dashboards built on Amazon QuickSight visualizing listing health and performance
- SQL-based analytics for querying operational and booking data
- Bookings data, ratings, reviews, and chat data all flowing into their internal system
- Market pricing data from tools like PriceLabs showing demand and pricing factors
- Properties being compared against benchmarks and comparable listings in the portfolio
- Regular on-ground property audit reports rating physical condition and readiness

SAY & DO

What does the stakeholder say and do? Any inconsistencies?

- "You cannot increase your supply within a day, it takes more than a few days to get a new property live"
- "I need to deep dive into the data, not just macro numbers"
- Manually cross-references lead-to-booking ratios to monitor demand vs supply
- Uses replacements and alternative unit offerings when a booked listing is unavailable
- Conducts regular property audits with on-ground teams to assess listing quality
- Actively building AI chat agents and dashboards to automate and improve operations
- Adjusts pricing and targets using tools like PriceLabs during peak seasons

PAINS	GAINS
<i>What are the biggest problems, frustrations, and challenges?</i> • Supply cannot be scaled quickly during peak season, getting a new property live takes days, leaving demand unmet • Lead-to-booking conversion funnel has poor performance; data exists but root cause is difficult to identify • Current data tools (PriceLabs, QuickSight) provide market-level data but lack granularity into the customer journey • At-risk listings are hard to detect proactively; issues often surface only after negative reviews accumulate • Macro-level data from third-party platforms does not go deep enough into guest experience or property owner patterns • Cancellation rates and audit findings require manual correlation, no automated at-risk flagging system	<i>What opportunities and benefits would make their job significantly easier?</i> • A real-time dashboard breaking down demand by area, unit type, size, and amenities requested by guests • More granular data capturing the full guest journey from inquiry to check-out for deeper insight • Automated alerts when listings show early at-risk signals (ratings drop, cancellations, audit failures) • AI-powered tools to automate guest communication and lead follow-up, improving conversion rates • Better lead-to-booking tracking to identify exactly where in the funnel prospects are dropping off • Deeper integration of property owner experience data to improve host supply management

Figure: Filled empathy map based on the domain expert interview with Mohammad Shahmeer, capturing his operational challenges, tools used, and desired BI capabilities.

3.3 Stage 2 - Define

Based on empathy stage insights, the following problem statement was formulated:

Problem Statement: A Regional Operations Manager from Airbnb needs a way to identify and predict geographic supply-demand gaps in real time because relying on lagging historical data makes it difficult to recruit hosts fast enough to meet sudden surges in guest interest.

Two 'How Might We' statements were developed to guide ideation:

- How might we create a predictive alerting system that identifies neighbourhoods where search volume is growing faster than active listing inventory, allowing the operational team to trigger host recruitment?
- How might we develop a unified market health dashboard that correlates host churn factors with guest demand to help the operational team decide where to allocate budget?

BI DESIGN SPRINT - DEFINE

*Design
Thinking
for BI*

So what's the business problem?

"[Stakeholder], a [role] from [company], needs a way to [need] because [insight]. Fulfilling this need will help the organization [business impact]."

Regional Operations Manager from Airbnb needs a way to identify and predict geographic supply-demand gaps in real time because relying on lagging historical data makes it difficult to recruit hosts fast enough to meet sudden surges in guest interests.

How might we... (1 or 2 statements - Focus on a clear BI problem)

- 1- How might we create a predictive alerting system that identifies neighborhoods where search volume is growing faster than active listing inventory allowing the operational team to trigger host recruitment
- 2- How might we develop a unified market health dashboard that correlates host churn factors with guest demand to help the operational team decide where to allocate budget.

Figure: Scanned page showing the problem statement and How Might We statements formulated during the design sprint.

3.4 Stage 3 - Ideate

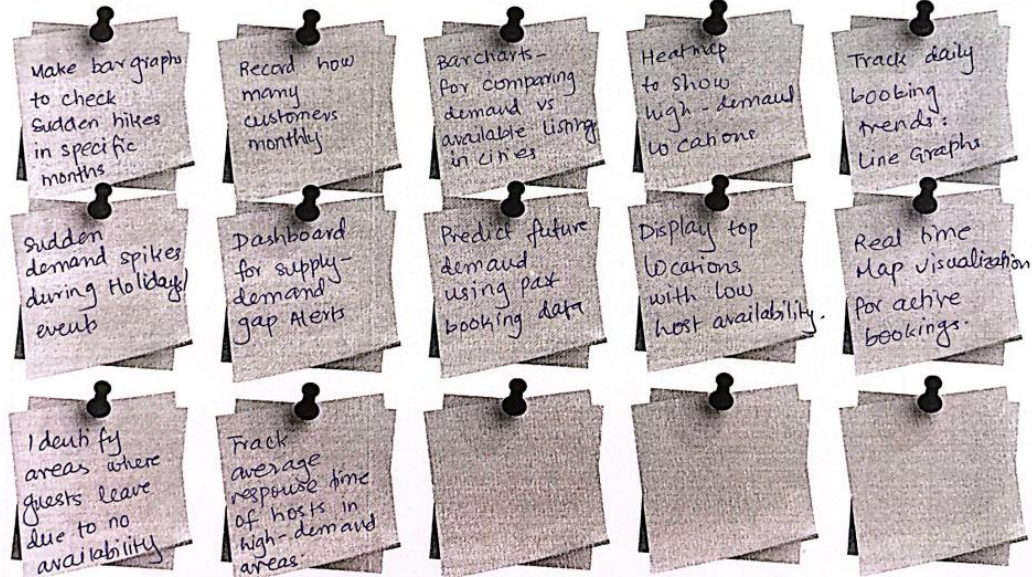
Each team member independently generated ideas using the sticky note method (one idea per note), following the principle of no judgement and high volume. A total of 20 ideas were generated across both team members, covering heat maps, funnel analysis, ML demand models, automated alerts, drill-down dashboards, competitor pricing integration, and real-time map visualizations.

BI DESIGN SPRINT - IDEATE

*Design
Thinking
for BI*

TEAM MEMBER 2 NAME Ayesha Salahuddin

Come up with as many ideas. One idea per sticky note.



No judgement - Go for volume - Be creative

Figure: Ayesha Salahuddin's ideation page showing 12 sticky note ideas including bar charts for demand spikes, real-time maps, supply-demand dashboards, and host availability tracking.

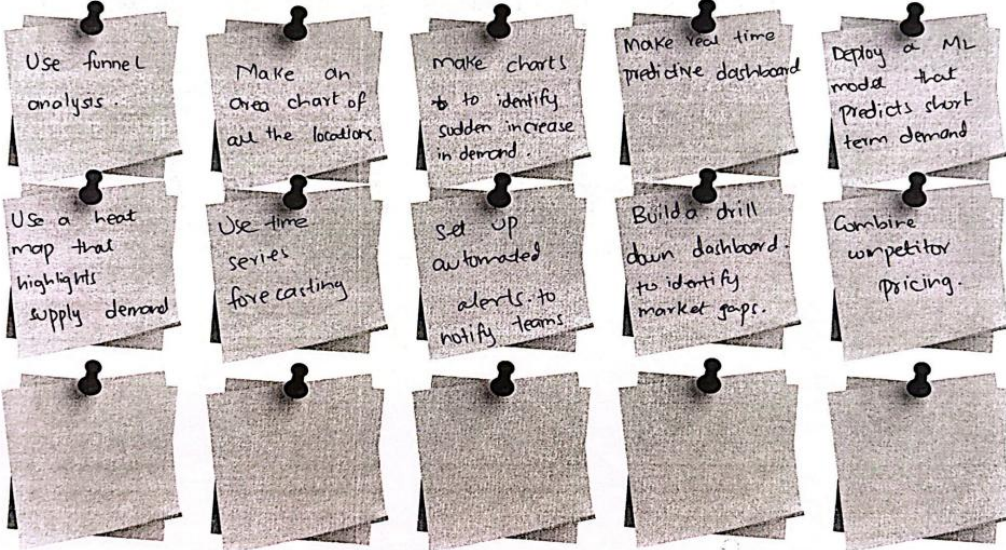
BI DESIGN SPRINT - IDEATE

*Design
Thinking
for BI*

TEAM MEMBER 1 NAME

Narin Nadeem

Come up with as many ideas. One idea per sticky note.



No judgement - Go for volume - Be creative

6

Figure: Narin Nadeem's ideation page showing 10 sticky note ideas including heat maps, funnel analysis, ML demand prediction, and automated alert systems.

3.5 Stage 4 - Prototype

The team combined the best individual Crazy 4s sketches into a group prototype, a hand-drawn wireframe of the Airbnb Supply-Demand Dashboard. This prototype directly guided the Power BI dashboard design.

BI DESIGN SPRINT - PROTOTYPE

Design
Thinking
for BI

Choose your favourite idea from the Crazy 4s. Make it as a group!

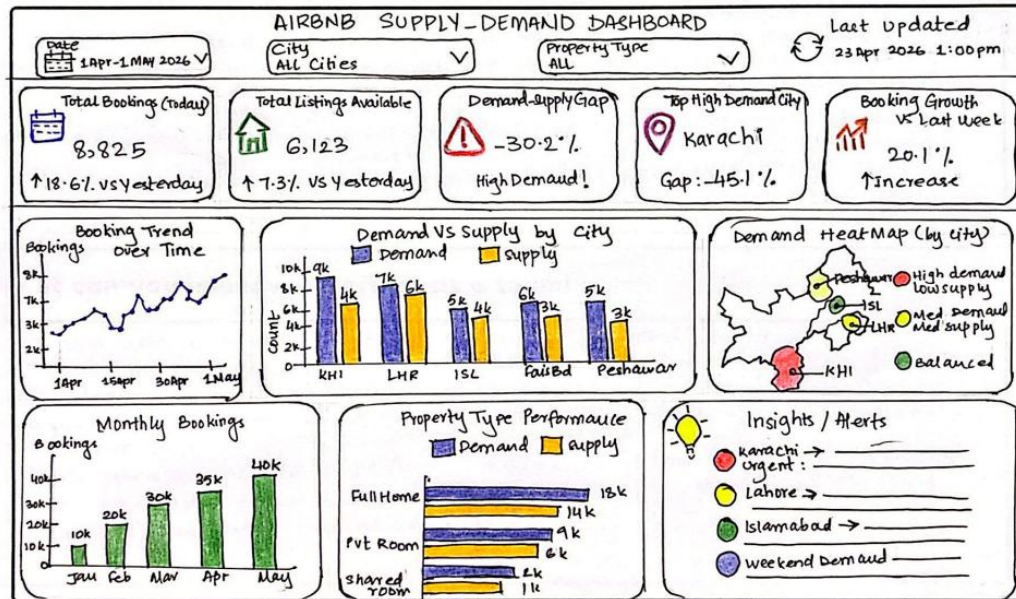


Figure: Hand-drawn group prototype of the Airbnb Supply-Demand Dashboard showing KPI cards, booking trend chart, demand vs supply bar chart, demand heat map by city, property type performance, and insights/alerts panel.

3.6 Stage 5 - Test

The prototype was tested against AI domain expert feedback, which provided three key insights: (1) shifting from vague alerts to specific actionable levers transforms the dashboard from a reporting tool into a playbook; (2) negative numbers in supply-demand gaps should function as visual alarms; (3) distinguishing between realized demand (bookings) and latent demand (searches) adds predictive capability. The team also reflected on four improvements: adding a Live vs Projected toggle, a weekday vs weekend filter, occupancy rates, and a price index.

BI DESIGN SPRINT - TEST

*Design
Thinking
for BI*

Upload the prototype picture & ask your AI expert for feedback!

Actionable Context: Your suggestion to shift from vague alerts (urgent) to specific levers (Host referral Bonus) is the "gold standard". It transforms the dash board from a reporting tool into a playbook.

The "Gap" Psychology: You're spot on about the colour coding. In operations, a negative number should not just be a digit; it should be a "visual alarm" that triggers a physical response.

Demand Definition: Distinguishing between Realized Demand (Bookings) and Latent Demand (Searches) adds a layer of sophistication that would allow an Ops Manager to Predict shortages before they actually happen.

What can you improve? Reflect as a team!

- We can add a "status: Live" vs "status: "Projected" toggle;
- We can improve our dashboard by adding a ~~like~~ the toggle stated above to clearly separate real time and fore-casted data and a Week-day vs weekend filter to understand supply gaps better (either they are due to gaps, pricing, availability)
- We can also include occupancy rates to focus more on demands
- We can add price index to evaluate prices and effects and over all customer demand better.

Figure: Scanned page showing AI expert feedback and team reflection on dashboard improvements.

4. Overall Analysis - BI Query Responses

4.1 BI Query 1: What is the supply-demand gap by NYC neighbourhood and borough?

Using availability 365 as the supply indicator and reviews per month as the demand proxy, a demand-supply gap score was calculated for each borough and neighbourhood. The analysis reveals that Manhattan and Brooklyn have the lowest average availability (141 and 129 days/year respectively) while maintaining high review rates indicating these boroughs are the most critically undersupplied relative to demand.

At the neighbourhood level, areas like Glen Oaks, Spring Creek, and Huguenot showed the highest gap scores, these are the priority areas for host recruitment campaigns. Staten Island and Bronx, while having higher average availability, showed lower review rates suggesting lower demand in these outer boroughs.

4.2 BI Query 2: How have booking trends changed over time?

Booking activity (proxied by last review date) shows a clear trend: steady growth from 2015–2019, a sharp crash in 2020 (COVID-19 pandemic impact), and a strong recovery through 2021–2022. Manhattan and Brooklyn show the most robust recovery, while Staten Island and Bronx lag behind. Entire home/apt listings recovered faster than shared rooms post-pandemic. Monthly data shows June and July as peak booking months, with a significant spike in July (1.11M reviews) indicating strong summer demand.

4.3 BI Query 3: Which listings are at-risk?

7,000 listings (6.9% of the dataset) were flagged as at-risk based on four criteria: reviews per month below 0.5, rating below 3.0, availability below 30 days, and no review in the past 180 days. Brooklyn has the highest concentration of at-risk listings (3,300) followed by Manhattan (3,000). Queens has 600 at-risk listings while Bronx and Staten Island have very few, suggesting the at-risk problem is concentrated in the high-supply boroughs. An additional 4,000 listings are classified as inactive (0 reviews, availability > 300 days).

4.4 BI Query 4: How does price vary across the dataset?

After cleaning, nightly prices range from \$50 to \$1,200 with an average of \$625.37. The price distribution is remarkably uniform across boroughs. Queens (\$631), Brooklyn (\$628), Bronx (\$627), Staten Island (\$624), and Manhattan (\$623) are all within \$8 of each other, suggesting that location does not drive price differentials as strongly as might be expected. By room type, Hotel rooms command the highest average price (\$668) followed by Shared rooms (\$633), Entire home/apt (\$625), and Private rooms (\$625). Pearson correlation confirms near-zero relationship between price and rating, hosts do not price based on quality scores.

4.5 BI Query 5: How does host concentration affect supply?

86.08% of listings are managed by individual hosts (fewer than 5 listings) while 13.92% are managed by commercial hosts (5+ listings). Commercial hosts tend to manage availability more strategically. The top 10 most expensive neighbourhoods (New Dorp: \$1,045, Chelsea Staten Island: \$1,042, Fort Wadsworth: \$1,024) are all in Staten Island, an unexpected finding suggesting niche premium markets exist outside the traditional Manhattan premium narrative.

5. Charts - Power BI Visualizations

The following charts were created in Microsoft Power BI Desktop using the cleaned NYC Airbnb dataset. Each chart addresses one or more of the five BI queries defined during the Design Thinking process.

5.1 Dashboard 1 - Supply-Demand Overview Charts

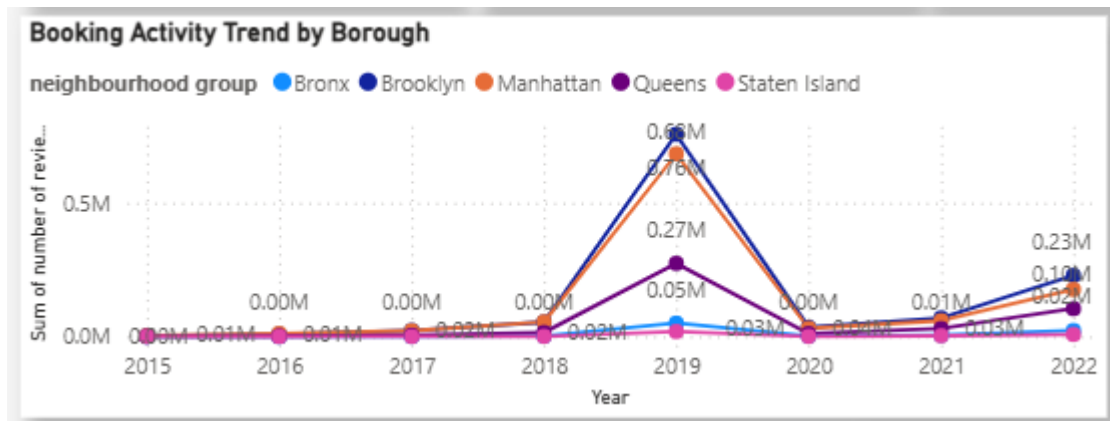


Figure: Line chart showing booking activity (review proxy) from 2015–2022 by borough. Reveals the 2020 COVID crash and 2021–2022 recovery, with Manhattan and Brooklyn showing the strongest rebound.

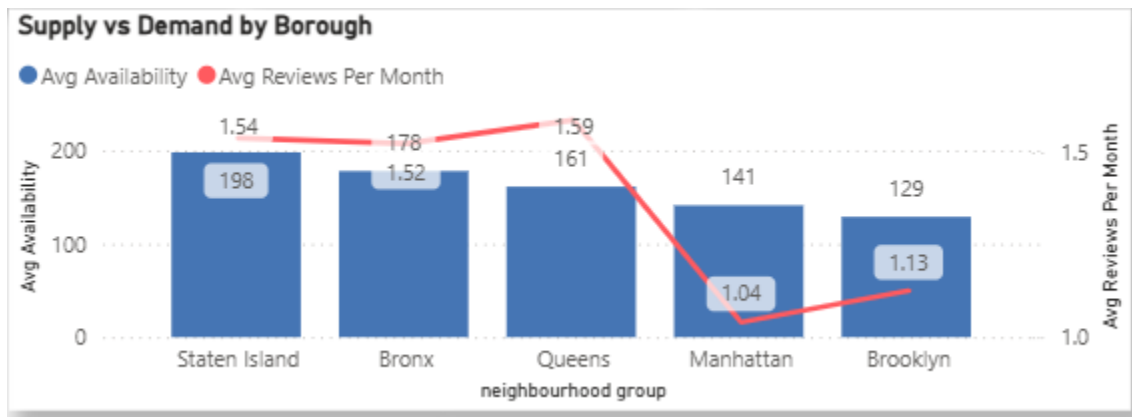


Figure: Dual-axis combination chart showing average availability (bars, supply) vs average reviews per month (line, demand) by borough. Manhattan and Brooklyn show the highest demand relative to supply, indicating critical undersupply.

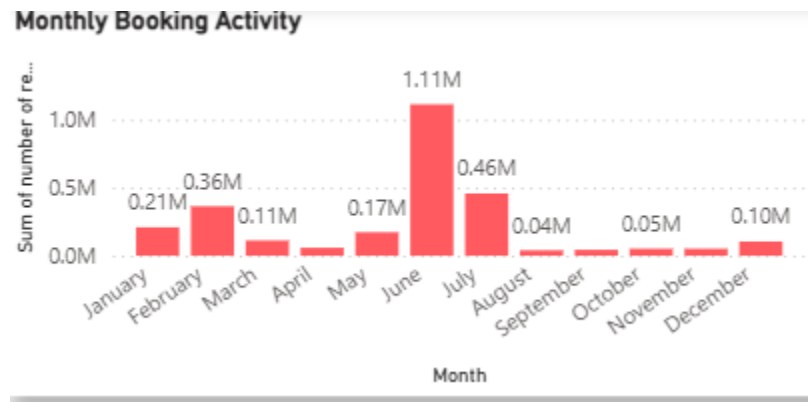


Figure: Column chart showing total review volume by month. June and July show peak demand (July: 1.11M), key months for proactive host recruitment campaigns.

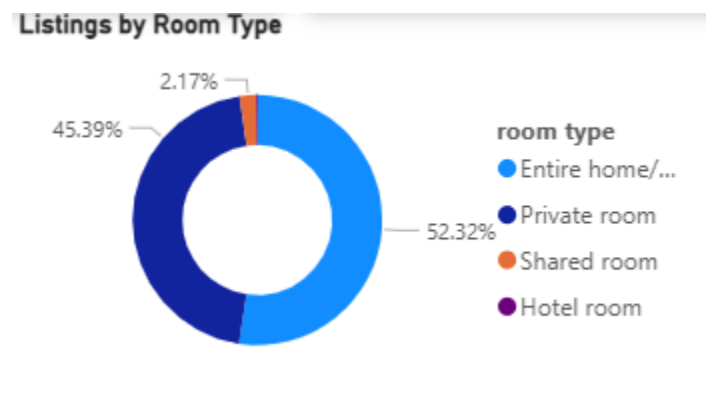


Figure: Donut chart showing listing share by room type. Entire home/apartment leads at 52.32%, followed by Private room at 45.39%, confirming that shared accommodation represents a small minority of NYC supply.

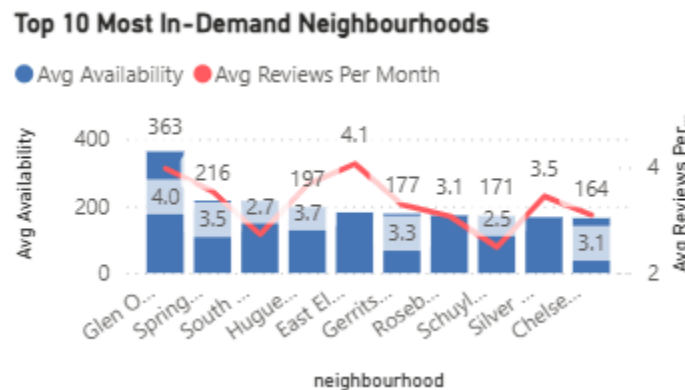


Figure: Combination chart showing average availability (bars) vs average reviews per month (line) for the top 10 highest-demand neighborhoods. Identifies specific areas where the gap between supply and demand is largest, actionable recruitment targets.

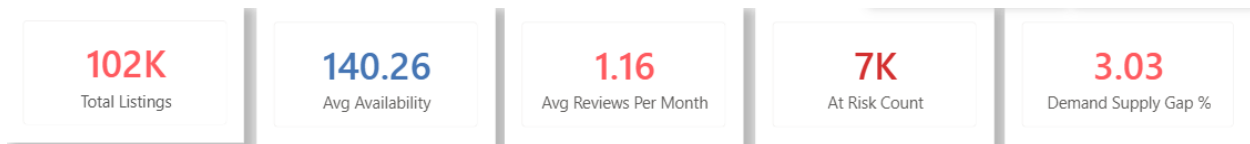


Figure: Five key performance indicators providing an at-a-glance snapshot of the NYC Airbnb market: 102K total listings, average availability of 140 days per year, 1.16 average reviews per month (demand proxy), 7K at-risk listings requiring urgent attention, and a demand-supply gap score of 3.03 indicating overall market undersupply.

5.2 Dashboard 2 - At-Risk & Quality Charts

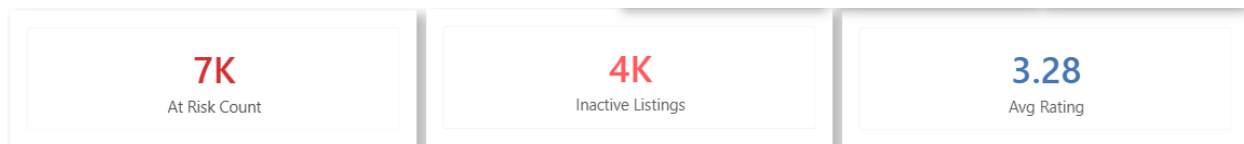


Figure: Three quality-focused KPIs showing that 7K listings are flagged as at-risk (declining reviews, low availability, rating below 3.0), 4K listings are completely inactive (zero reviews, availability over 300 days), and the overall average review rating across all NYC listings is 3.28 out of 5 indicating moderate market quality with significant room for improvement.

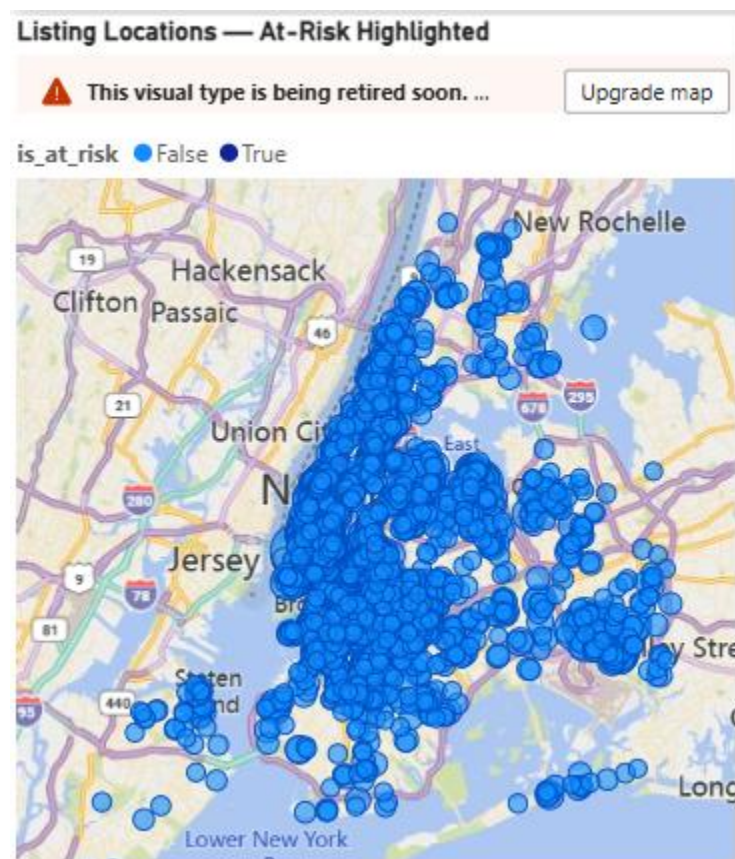


Figure: Geographic map of NYC showing all listing locations. Blue dots = safe listings, red dots = at-risk listings. Concentration of listings visible in Manhattan, Brooklyn, and Queens with at-risk listings visible as a distinct layer.

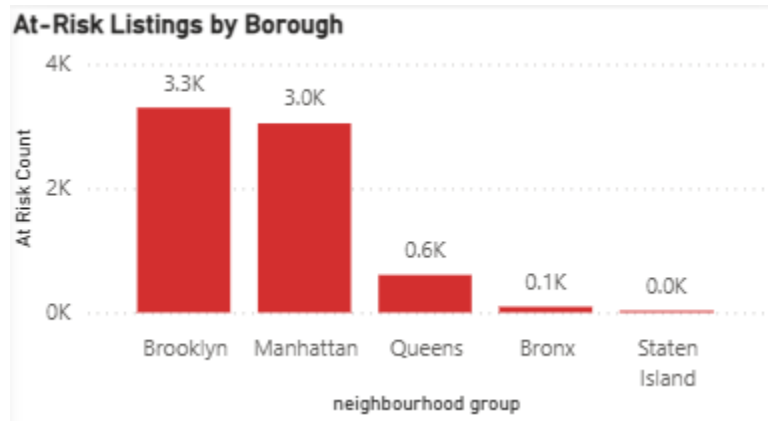


Figure: Column chart showing count of at-risk listings by borough. Brooklyn leads with 3,300 at-risk listings, followed by Manhattan with 3,000, these boroughs require urgent intervention to prevent market quality degradation.

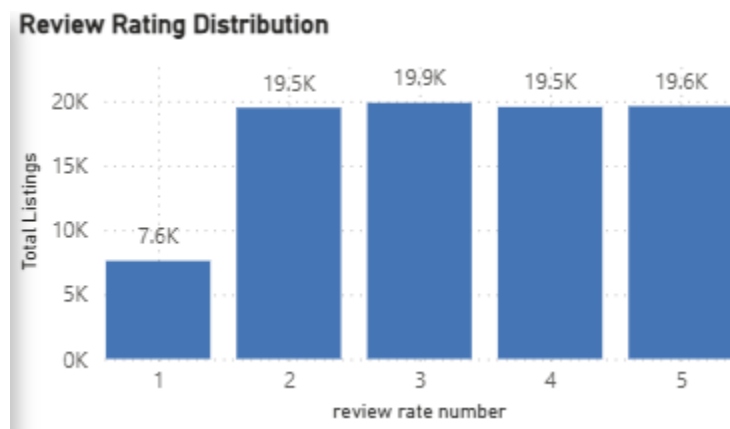


Figure: Column chart showing number of listings at each rating level (1–5). Ratings 2–5 are relatively evenly distributed (~19,500–19,900 listings each) while rating 1 has only 7,600 listings, most listings maintain at least moderate quality.

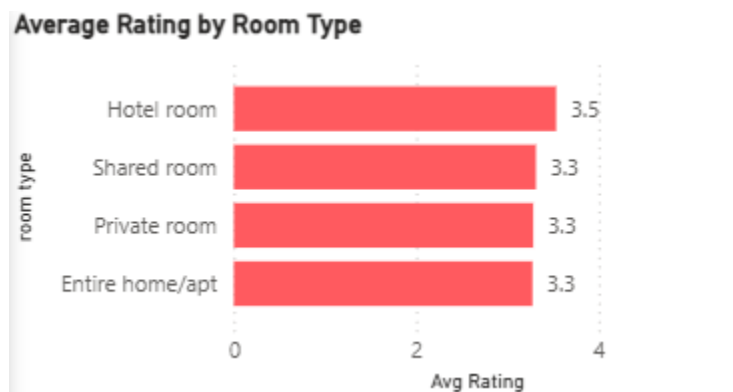


Figure: Horizontal bar chart comparing average review rating by room type. Hotel rooms lead with 3.5, while Private rooms, Shared rooms, and Entire home/apt all average 3.3, minimal variation across room types.

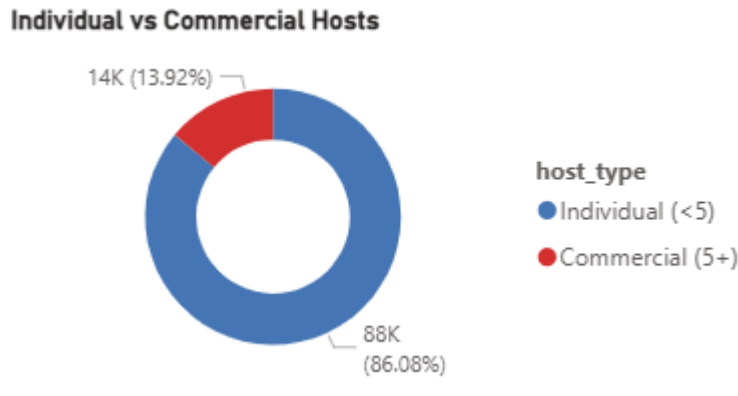


Figure: Donut chart showing host type split. Individual hosts (<5 listings) make up 86.08% of all listings while commercial hosts (5+ listings) account for 13.92%, the market is predominantly driven by individual hosts.

5.3 Dashboard 3 - Pricing & Revenue Charts

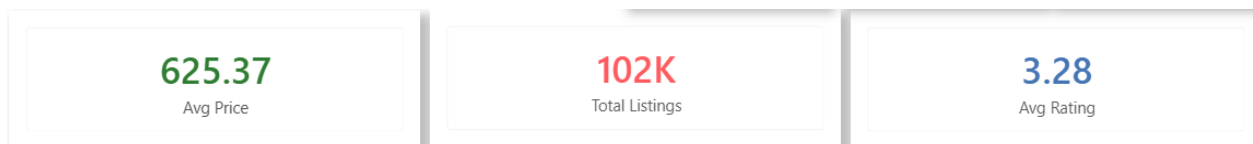


Figure: pricing-focused KPIs showing an average nightly price of \$625.37 across all NYC listings, 102K total listings in the cleaned dataset, and an average review rating of 3.28 providing a quick baseline for pricing benchmarks before drilling into borough, room type, and neighbourhood-level price variations.

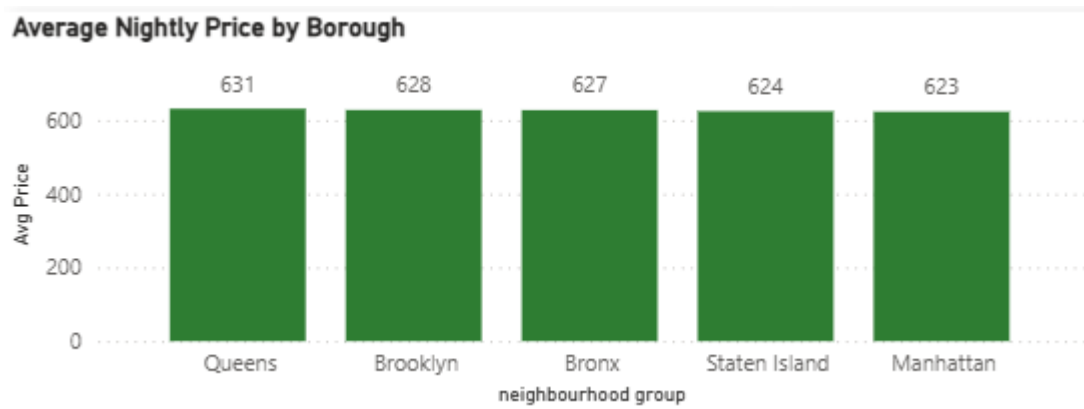


Figure: Column chart showing average nightly price by borough. Prices are remarkably uniform: Queens (\$631), Brooklyn (\$628), Bronx (\$627), Staten Island (\$624), Manhattan (\$623), location alone does not drive significant price premiums in this dataset.



Figure: Horizontal bar chart showing average price by room type. Hotel rooms command the highest average (\$668), followed by Shared rooms (\$633), with Entire home/apt and Private rooms both at \$625.

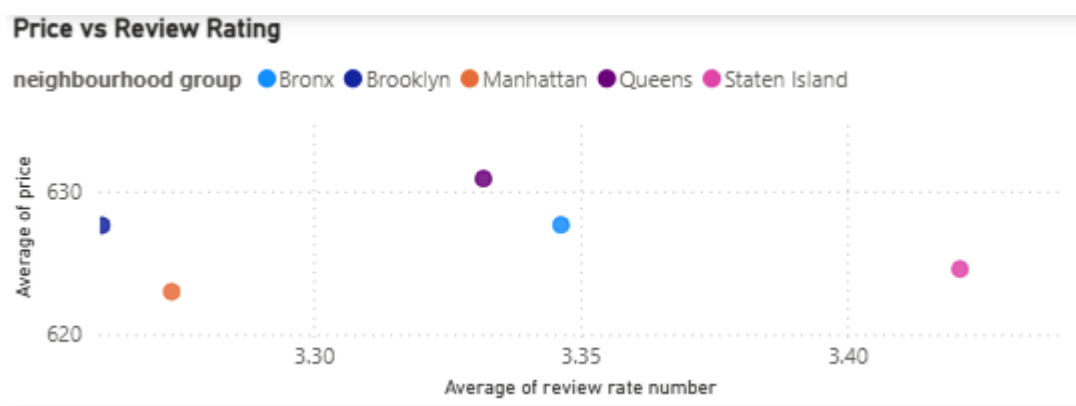


Figure: Scatter plot showing average price vs average review rating by borough. Confirms near-zero correlation, higher-priced listings do not consistently receive better ratings, and vice versa.



Figure: Horizontal bar chart showing the 10 highest-priced neighbourhoods. New Dorp (\$1,045), Chelsea Staten Island (\$1,042), and Fort Wadsworth (\$1,024), all in Staten Island lead the rankings, revealing an unexpected premium market outside Manhattan.

6. Dashboards - Power BI

6.1 Dashboard 1 - Supply-Demand Overview

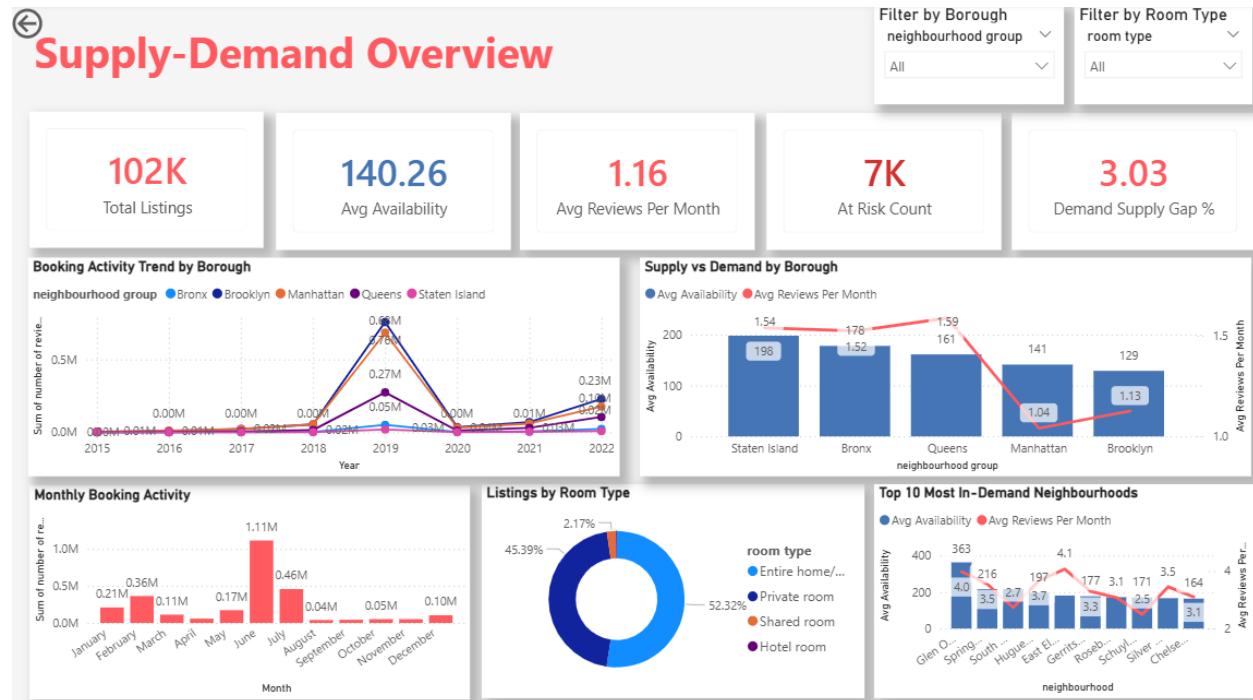


Figure: The primary operational dashboard showing 5 KPI cards (Total Listings: 102K, Avg Availability: 140 days, Avg Reviews/Month: 1.16, At-Risk Count: 7K, Demand-Supply Gap: 3.03), booking trend line chart, supply vs demand dual-axis chart, monthly bookings column chart, room type donut chart, and top 10 in-demand neighbourhoods chart. Allows filtering by borough and room type.

6.2 Dashboard 2 - At-Risk & Quality Analysis

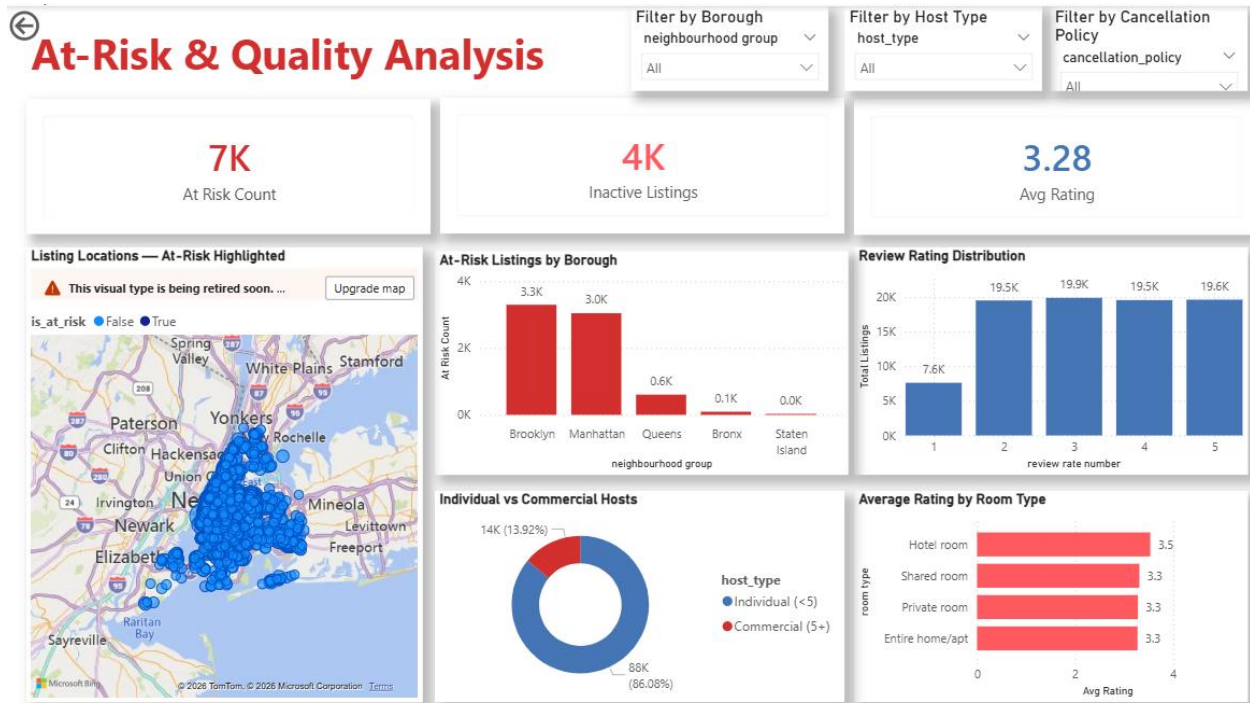


Figure: Quality monitoring dashboard showing KPI cards (At-Risk: 7K, Inactive: 4K, Avg Rating: 3.28), geographic map with at-risk listings highlighted, at-risk count by borough, review rating distribution, average rating by room type, and individual vs commercial host split. Allows filtering by borough, host type, and cancellation policy.

6.3 Dashboard 3 - Pricing & Revenue Analysis

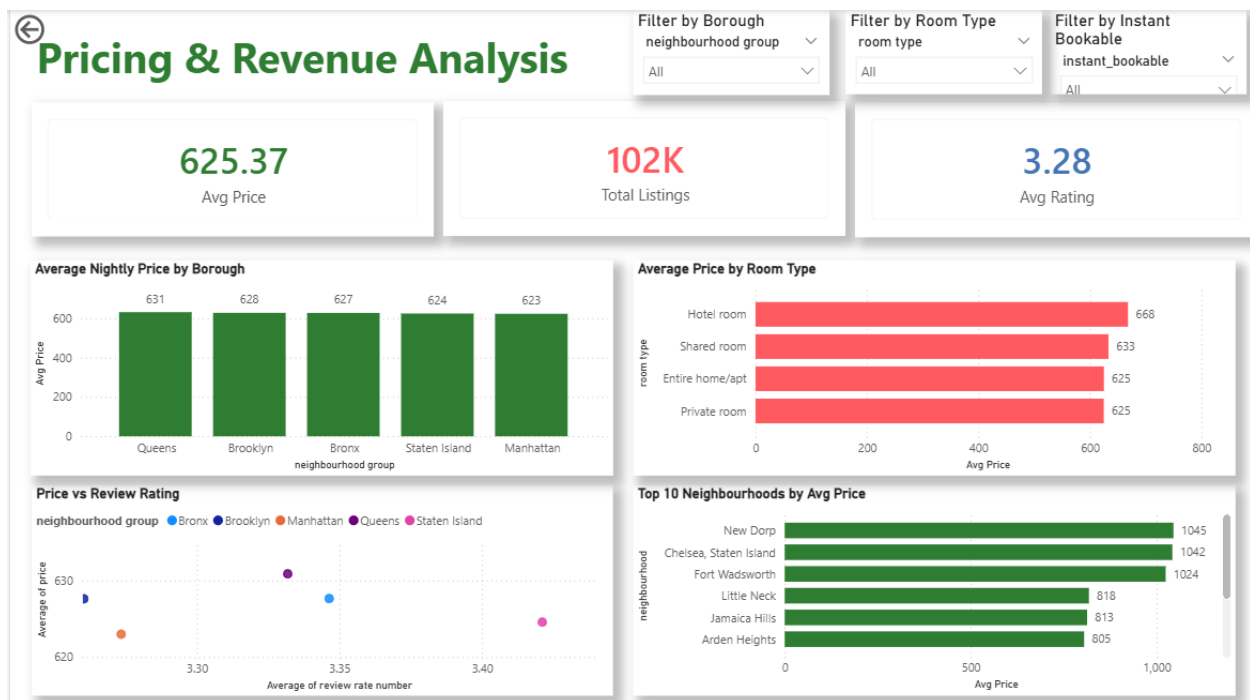


Figure: Pricing intelligence dashboard showing KPI cards (Avg Price: \$625.37, Total Listings: 102K, Avg Rating: 3.28), average price by borough, average price by room type, price vs rating scatter plot, and top 10 highest-priced neighbourhoods. Allows filtering by borough, room type, and instant bookability.

7. Data Story

7.1 Story Overview

The data story presents the NYC Airbnb supply-demand problem as a four-step narrative, moving from identifying the problem to delivering an actionable solution. The story is designed to be walked through sequentially by the Regional Operations Manager to understand the full picture before making decisions.

7.2 The Story - Step by Step

Step 1: The Problem

NYC Airbnb has over 102,000 listings, but 7,000 are at-risk and 4,000 are completely inactive. The operations team relies on weekly reports, by the time they identify a problem and act on it, the demand surge has already passed and revenue has been lost.

Step 2: The Evidence

Booking activity peaked in 2019 then crashed in 2020 due to COVID-19. Manhattan and Brooklyn, the two highest-demand boroughs have the lowest average availability (129–141 days/year), creating a critical supply-demand mismatch. The monthly data confirms that June and July are the most critical months requiring proactive supply management.

Step 3: The Risk

Brooklyn has 3,300 at-risk listings and Manhattan has 3,000. These listings are showing declining review rates, low availability, and ratings below 3.0. If left unaddressed, they will damage the overall market quality score for these boroughs, reducing guest trust and long-term platform revenue.

Step 4: The Solution

The solution is to target host recruitment campaigns in high-gap neighbourhoods like Glen Oaks and Spring Creek, where demand (reviews per month) significantly outpaces available supply. Hotel rooms command the highest average price (\$668), and instant bookable listings drive more demand, these insights guide both pricing and listing type recruitment strategies. The real-time dashboard with automated alerts enables the operations team to act within hours rather than days.



The NYC Airbnb Supply-Demand Story

STEP 1 - THE PROBLEM

NYC Airbnb has 102,000+ listings but 7,000 are at-risk and 4,000 are completely inactive.

The operations team relies on weekly reports by the time they act, demand has already passed.

STEP 2 - THE EVIDENCE

Booking activity peaked in 2019 then crashed in 2020 (COVID).

Manhattan and Brooklyn have the lowest availability (129-141 days) but the highest demand, creating a critical supply gap.

STEP 3 - THE RISK

Brooklyn has 3,300 at-risk listings. Manhattan has 3,000.

These listings have declining reviews, low availability, and ratings below 3.0, they will hurt overall market quality if not addressed as soon as possible.

STEP 4 - THE SOLUTION

Target host recruitment in high-gap neighborhoods like Glen Oaks and Spring Creek.

Hotel rooms command the highest price (\$668 avg). Instant bookable listings drive more demand.

Real-time alerts + this dashboard

This dashboard empowers the Regional Operations Manager to move from reactive to proactive decision-making, identifying supply gaps before they become lost revenue, and flagging at-risk listings before ratings drop.

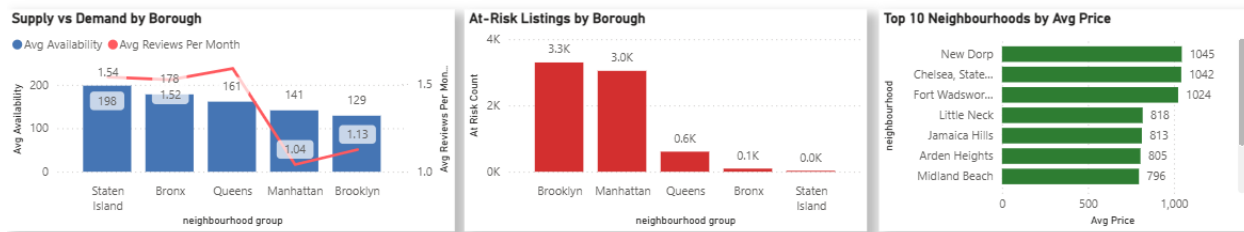


Figure: The complete data story page in Power BI showing all four story steps (Problem, Evidence, Risk, Solution) as colour-coded narrative cards, with three supporting charts below and a concluding statement explaining how the BI solution transforms reactive operations into proactive decision-making.

7.3 How the Story Solves the Problem

This data story directly addresses the core business problem identified during the Design Thinking process. By guiding the Regional Operations Manager through a sequential narrative, from recognizing the supply-demand gap, to understanding its geographic and temporal patterns, to identifying at-risk listings before they damage market quality, to taking specific recruitment and pricing actions, the story transforms raw data into a decision-making playbook.

The three supporting charts on the data story page (Supply vs Demand by Borough, At-Risk by Borough, Top 10 Neighbourhoods by Price) provide the evidence base for each recommended action, ensuring that every decision is grounded in data rather than intuition. This directly addresses the stakeholder pain point identified in the empathy map: 'By the time I see data, it's too late.'

8. Work Contributions

Ayesha Salahuddin 29157

Ayesha led the data wrangling and exploratory data analysis work, writing the Python notebook covering data validation, all 10 cleaning steps, descriptive statistics, and 10 EDA visualizations with interpretations. She also conducted the three statistical tests while also co-designing the group prototype dashboard sketch. She was responsible for conducting the business professional interview with Mohammad Shahmeer and managing all interview documentation including the recording, empathy map, and interviewee details submission. Ayesha also took the lead on the Power BI dashboard development building all three interactive dashboards (Supply-Demand Overview, At-Risk & Quality, Pricing & Revenue) and the data story page. She handled dashboard formatting, colour coding, slicer configuration, and DAX measure creation. Additionally, Ayesha contributed to the report document, writing the dashboard descriptions, chart captions, EDA summary and data story section.

Narin Nadeem 29167

Narin led the ideation and prototyping stages of the Design Thinking sprint, contributing to the sticky note ideas during the Ideate phase. Narin was responsible for completing the Background Data Knowledge Excel file filling all four sheets (Business Knowledge, Competitor Analysis, Pain Areas, and Data Info) with accurate information derived from the actual dataset and the understanding of the dataset and background knowledge. She also co-designed the Crazy 4s individual sketch. Narin also drafted the problem statement and How Might We statement during the Define stage. Narin also helped on the Power BI dashboard development building all three interactive dashboards.

She also contributed to dashboard formatting, colour coding, slicer configuration, and DAX measure creation with Ayesha. Additionally, she contributed to the report by writing the dataset description, industry background, EDA summary, and design thinking implementation sections.

NOTE: Both team members collaborated equally on the overall project direction, Design Thinking sprint completion, review of all deliverables, and final report compilation. All major decisions including the choice of BI queries, dashboard structure, and data story narrative were made jointly.